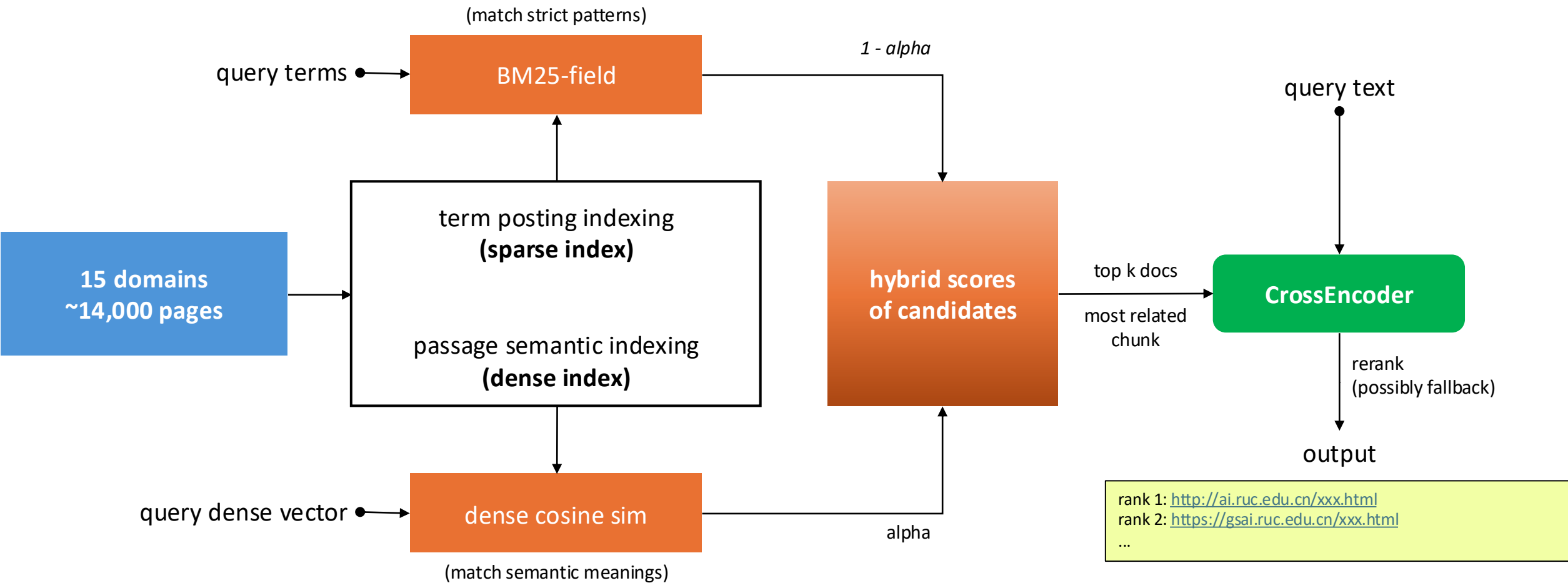
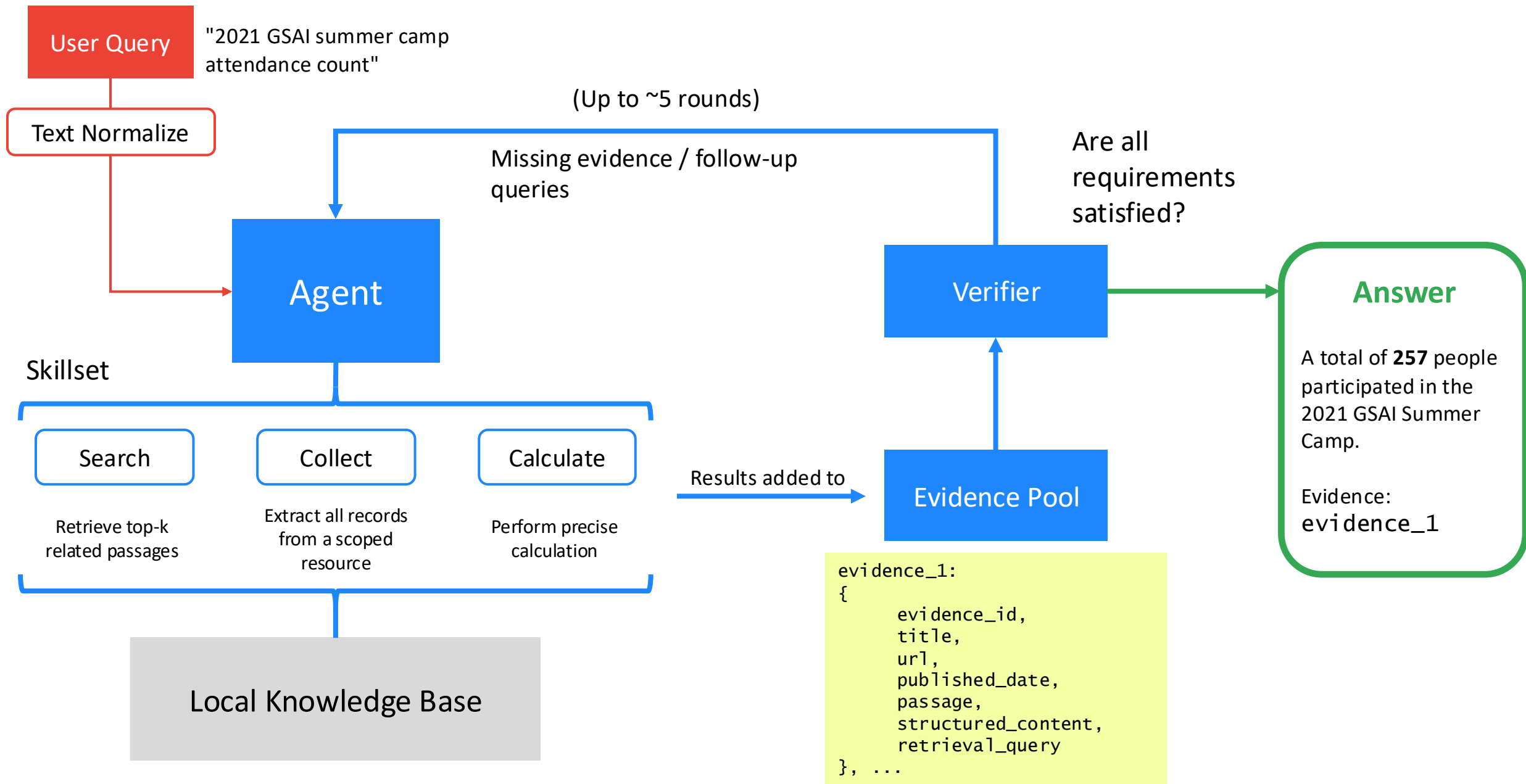


zeta-engine

面向中国人民大学站点的搜索引擎与 RAG 增强检索

高瓴人工智能学院 熊洋铭





Request routing

Interfaces

$$r_{\text{CLI/HTTP}} \xrightarrow{\text{Parse}} (o, \theta) \xrightarrow{\text{Dispatch}} y \xrightarrow{\text{Format}} r_{\text{out}}.$$

Parse the operation and arguments, dispatch the request, format the result.

Application

$$(q, m) \xrightarrow{\text{Normalize}} (\tilde{q}, m),$$
$$\text{Service}(\tilde{q}, m) = \begin{cases} \text{SearchDocuments}(\tilde{q}), & m = \text{search}, \\ \text{RAG}(\tilde{q}; f_s, f_c), & m = \text{rag}. \end{cases}$$

Search returns documents. RAG receives search and collection callbacks.

Data preparation

Ingestion

$$\begin{aligned} U_{\text{seed}} &\xrightarrow{\text{NormalizeURL+ScopeCheck}} Q \xrightarrow{\text{Schedule+Fetch}} H, \\ H &\xrightarrow{\text{Extract+Clean}} \mathcal{D}, \\ d &= (\text{URL}, \text{title}, \text{text}, h_{\text{sem}}), \quad d \in \mathcal{D}. \end{aligned}$$

Q: persistent crawl queue H: downloaded HTML D: document corpus

Infrastructure

$$\begin{aligned} x &\xrightarrow{\text{NFKC+Casefold+Whitespace}} \tilde{x} \xrightarrow{\text{Tokenize}} \{(w_i, p_i)\}, \\ B_t &\xrightarrow{\text{Write}(\Delta)} B_{t+1}, \quad (B_t, k) \xrightarrow{\text{Read}} v_k. \end{aligned}$$

Normalize and tokenize text. Read and write document, queue and index records.

Retrieval and evaluation

Retrieval

$$\begin{aligned}\mathcal{D} &\xrightarrow{\text{Index+Chunk+Encode}} (I_{\text{BM25F}}, I_{\text{FTS5}}, V), \\ \tilde{q} &\xrightarrow{\text{Sparse} \cup \text{Dense}} C \xrightarrow{(1-\alpha)\tilde{s}_s + \alpha\tilde{s}_d} C' \xrightarrow{\text{CrossEncoder}} R_k, \\ R_k &\xrightarrow{\text{Output}} \begin{cases} \text{DocumentCards}(R_k), & \text{search,} \\ \text{RecoverStructure}(R_k, \mathcal{D}), & \text{rag.} \end{cases}\end{aligned}$$

Fuse by document for search and by passage for RAG, then rerank.

Evaluation

$$\begin{aligned}\{q_i\}_{i=1}^N &\xrightarrow{\text{Search/RAG}} \{y_i\}_{i=1}^N \xrightarrow{\text{Submit+Score}} \{s_i, \ell_i\}_{i=1}^N, \\ \{s_i, \ell_i\}_{i=1}^N &\xrightarrow{\text{Aggregate}} \text{EvaluationReport}.\end{aligned}$$

Run queries, submit results and aggregate scores with latency.

Agentic RAG

State, tool actions and verification

$$\begin{aligned} S_0 &= (\tilde{q}, E_0 = \emptyset, L_0 = \emptyset), \\ a_t &= \text{Agent}(S_t), \\ S_{t+1} &= \begin{cases} \text{Update}(S_t, \text{Tool}(a_t)), & a_t \in \mathcal{T}, \\ \text{Update}(S_t, \text{Verifier}(A_t, E_t)), & a_t = \text{Answer}, \end{cases} \\ S_T &\xrightarrow{\text{Finalize}} (\text{Answer}, \text{Sources}, \text{Status}), \\ \mathcal{T} &= \{\text{Search}, \text{Collect}, \text{Calculate}\}. \end{aligned}$$

S: queries, evidence, calculations and verification feedback

Tools update the state. Verifier checks the candidate answer.

Up to 5 rounds and 12 model calls. The call limit terminates with an error.

Complete evidence for reliable counts

Search

Partial matches

ID	Record
3	C
4	D

Relevant, but incomplete



Collection

Recover the full table

ID	Record
1	A
2	B
3	C
4	D
5	E



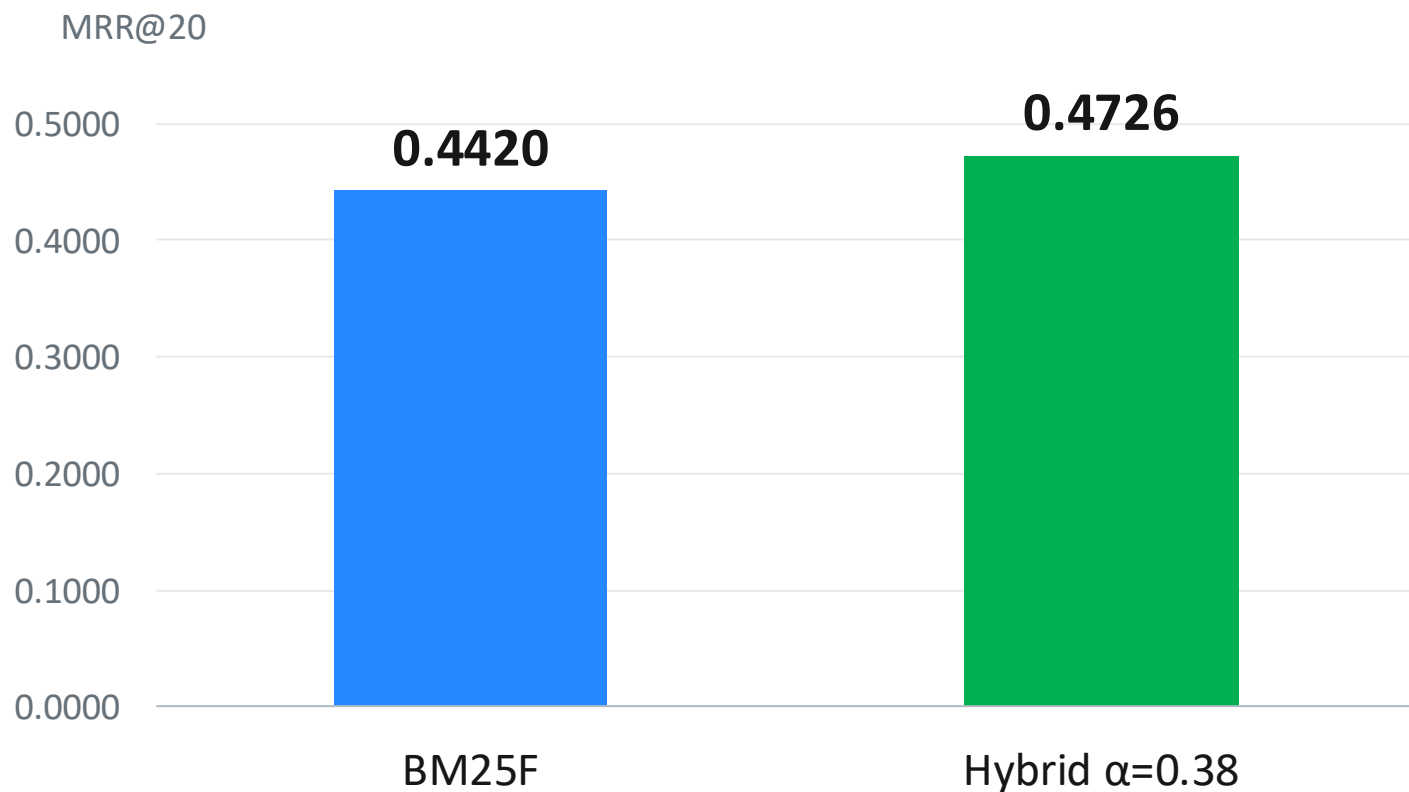
Calculate

Deterministic counting

5 records

Source: evidence_T1

Hybrid retrieval improves ranking



+6.9%

Relative MRR@20 gain

25 → 26

Top-1 hits

Recall@20 stays at 0.7361

One interface for search and questions



01 Ask a question

Policies and notices

02 Check sources

Review source pages
and evidence support

> `uv run zeta-engine serve`